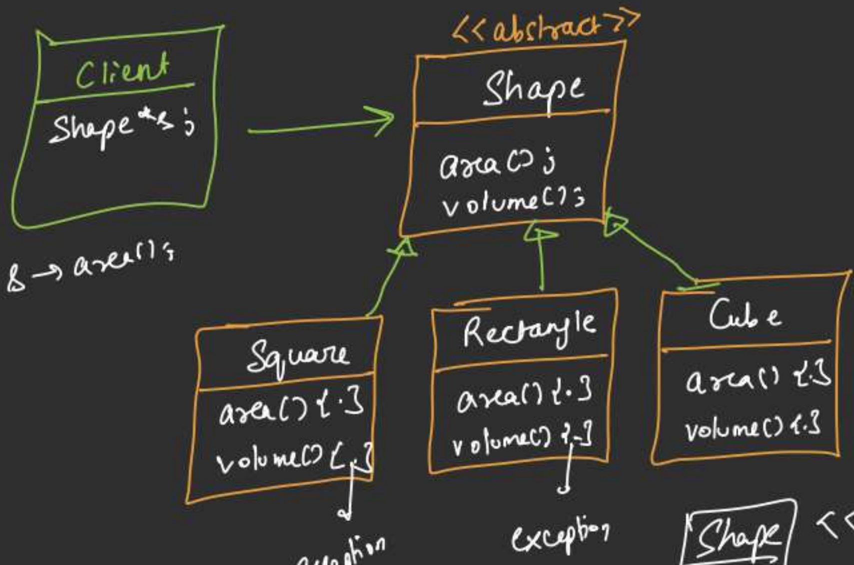
SOL[ID]

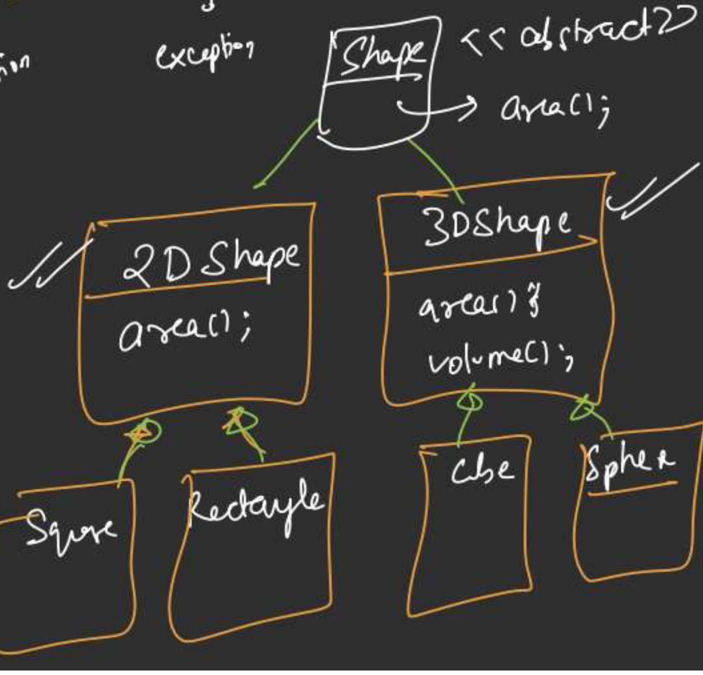
Class Diagram ⇒ Code

⇒ (A*)

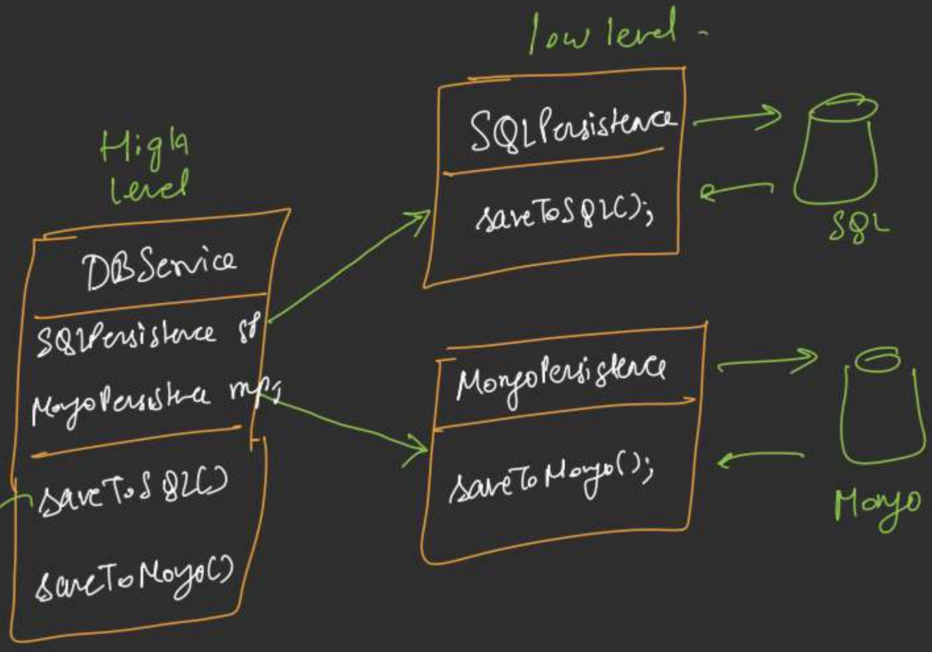
⇒ Haversine

I : Interface Segregation principle.

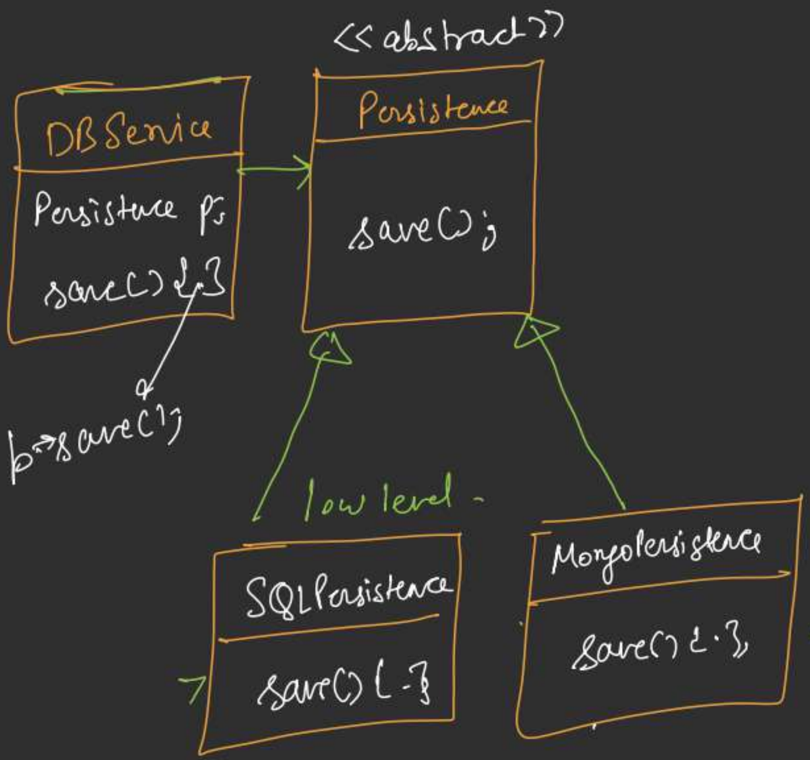




D: Dependency Inversion Principle




```
(){
  sp -> saveToSQL();
}
```



SOLID

Singleton
Strategy
Factory

Singleton

```
Class A {
    f1() {..}
    f2() {..}
};
```

new

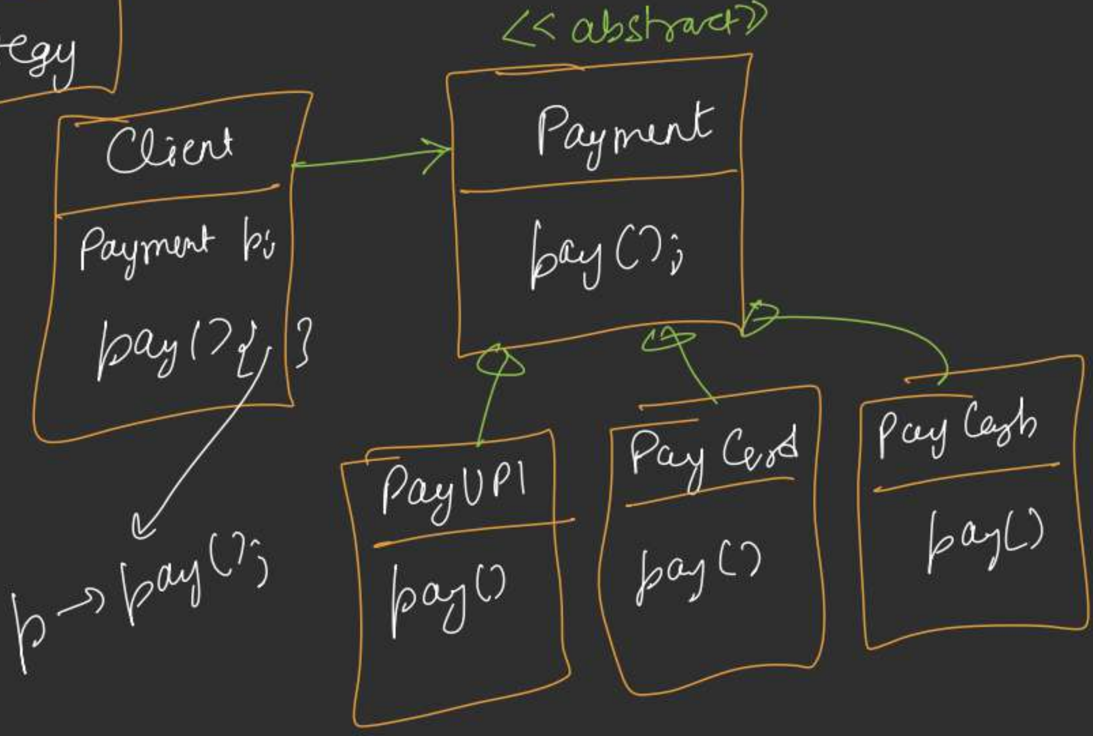
new A()

```
Class A {
  Private:
    A() {..}
    A *a = NULL;
  Public:          -> static
    getObject() {
      if(a == NULL) {
        a = new A();
        return a;
      }
      return a;
    }
}
```

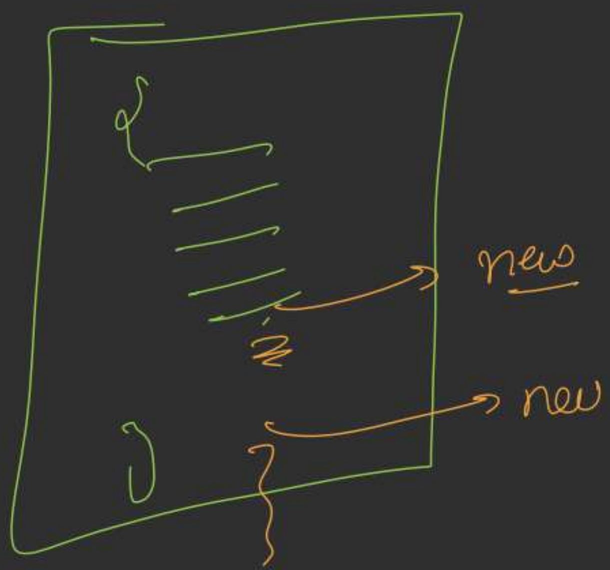
A * a1 = A :: getObject();

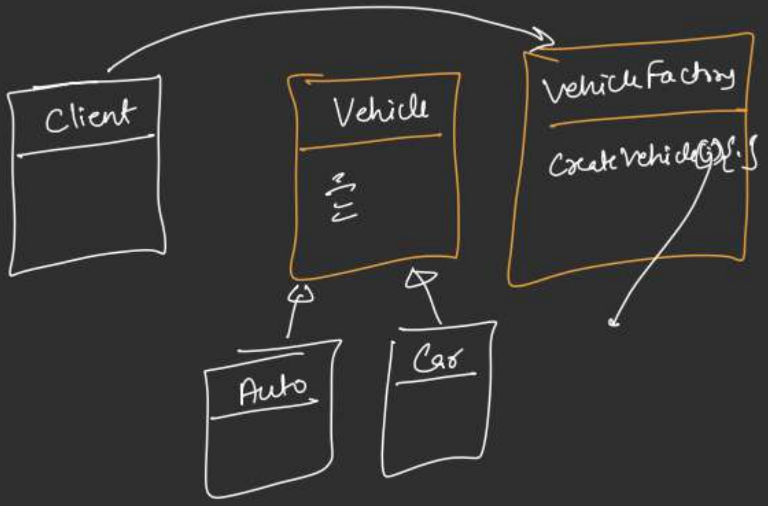
A * a2 = A :: getObject();

Strategy



Factory → Object creation



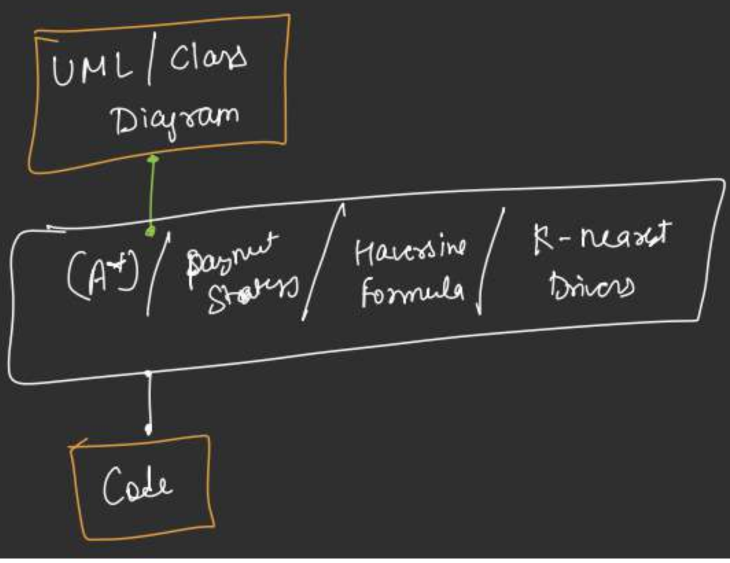



```
Vehicle*
createVehicle (String type){
    if( type == `Auto")
        return new Auto();
    else if( type == `Car'){
        return new Car();
    }
}
```

SOLID

+

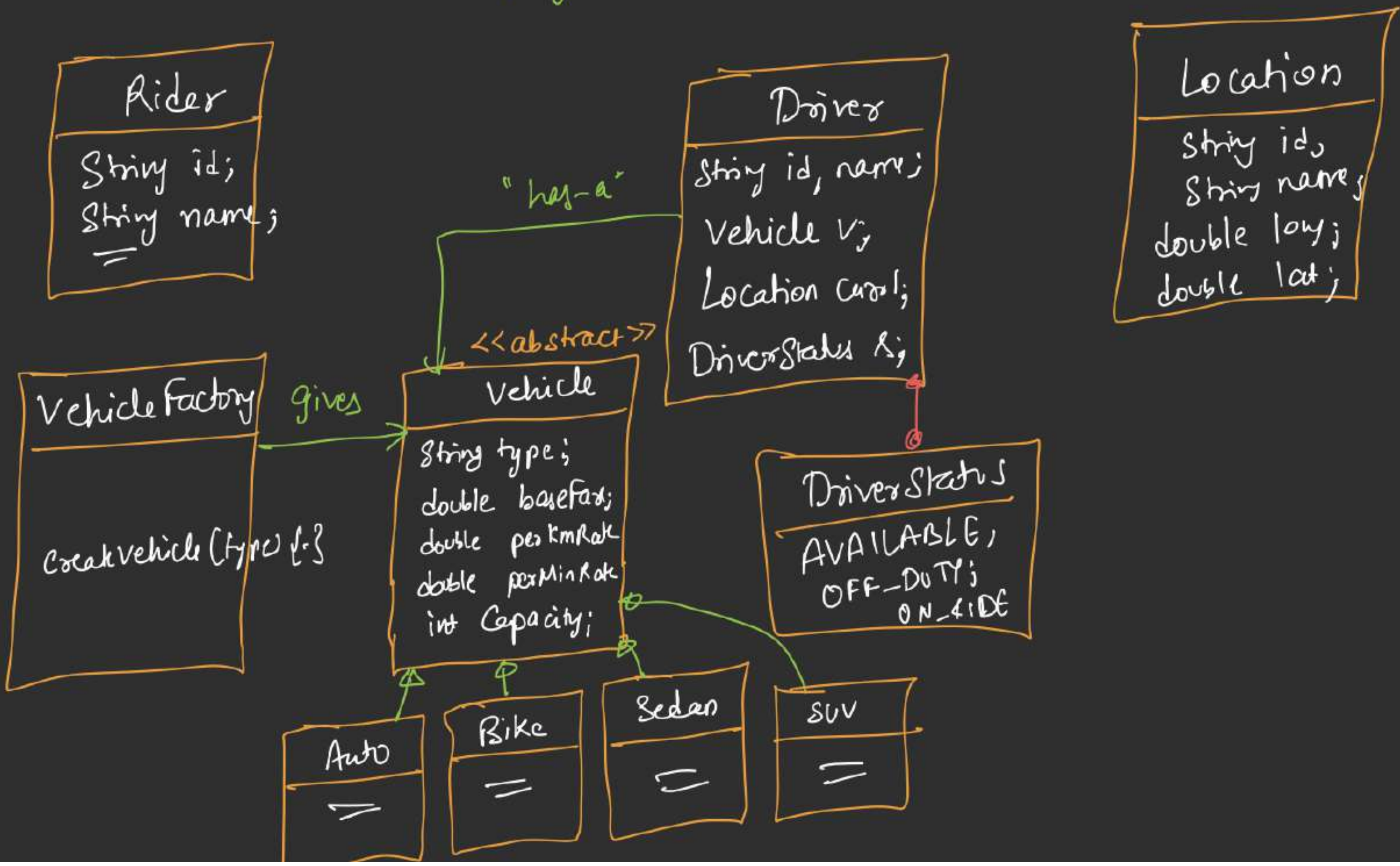
Strategy + Factory + Singleton

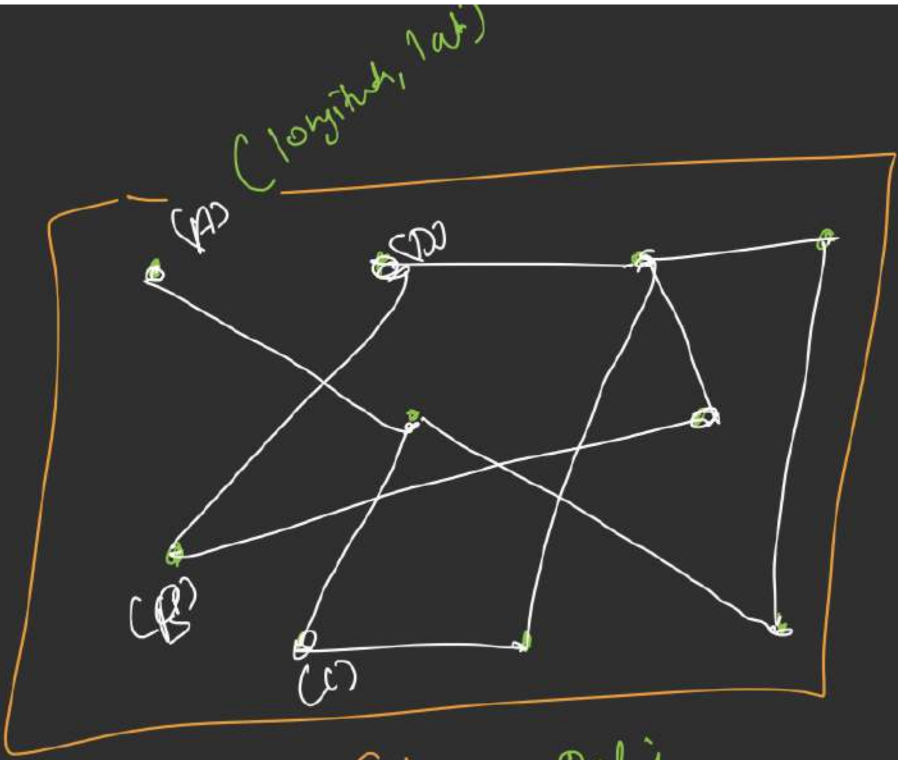


Objects (UBER - Project)

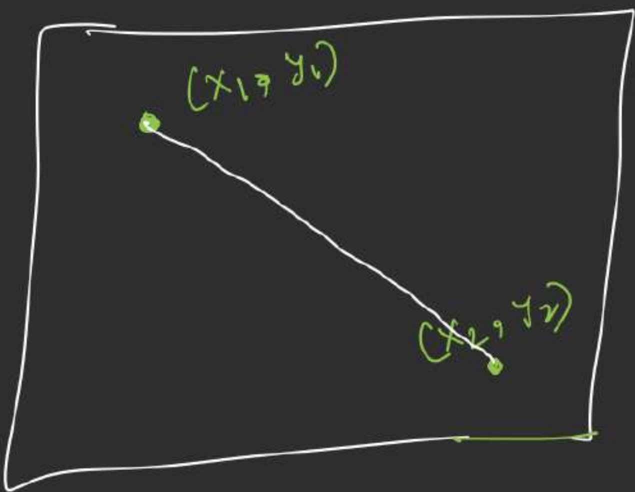
- Driver ⇄ DriverMatchService
- Rider
- Ride ⇄ Ride Service
- Vehicle (Auto/ Car/ Bike)
- CityGraph. ⇄ RouteService
  - Edge
- Location

Payment.





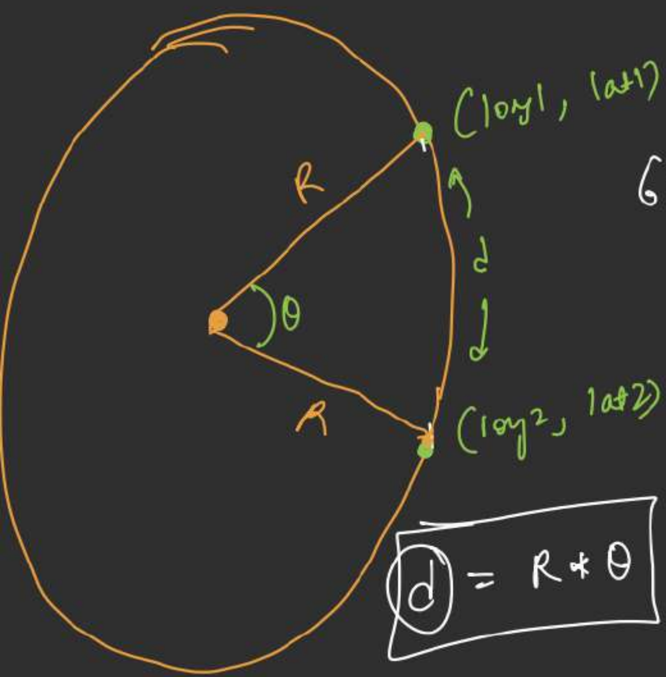
City. → Delhi



Haversine

Dist. Formula

↳ $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$



$$\boxed{d = R * \theta}$$

67

$$2\pi r = 360^\circ$$

$$2\pi * r = 2\pi r$$

$$360^\circ = 2\pi$$

$$180^\circ = \pi$$

$$\boxed{1^\circ = \frac{\pi}{180}}$$